

Why is $\cos 2x = 2\cos^2 x - 1$?

From the angles formula:

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\text{Set } A = B = x$$

$$\cos(x+x) = \cos x \cos x - \sin x \sin x$$

$$\boxed{\cos(2x) = \cos^2 x - \sin^2 x}$$

Use the fundamental identity:

$$\underline{\sin^2 x + \cos^2 x = 1}$$

$$\Rightarrow \sin^2 x = 1 - \cos^2 x$$

$$\Rightarrow \cos^2 x - (1 - \cos^2 x)$$

$$\Rightarrow \cos^2 x - 1 + \cos^2 x$$

$$\boxed{\Rightarrow 2\cos^2 x - 1}$$